

Exam Date & Time: 16-Nov-2025 (09:30 AM - 12:30 PM)



# MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU  
SEVENTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

## INTRODUCTION TO COMMUNICATION SYSTEMS [ECE\_4313]

**Marks: 50**

**Duration: 180 mins.**

### PART-A

**Answer all the questions.**

- 1) Apply the Nyquist Sampling Theorem to explain how it is used in converting an analog audio signal into its digital form, and demonstrate its significance in real-world audio recording systems. (5)
  - A)
  - B) Explain how data rate is influenced by available bandwidth. (3)
  - C) Illustrate the relationship between modulation technique and required bandwidth (2)
- 2) Illustrate the structure and working principle of an optical fibre, and analyze its advantages over conventional transmission media. (5)
  - A)
  - B) Apply the concept of Wavelength Division Multiplexing (WDM) to explain how multiple signals can be transmitted simultaneously through a single fibre. (3)
  - C) Interpret one major benefit and one challenge of optical communication in modern networks. (2)
- 3) Differentiate between direct broadcast satellite and digital satellite radio in terms of coverage and usage. (5)
  - A)
  - B) Apply how data analytics differs from cloud computing in function and purpose. (3)
  - C) Interpret the unique advantage of ultrawideband (UWB) compared to other wireless communication technologies. (2)
- 4) Explain the concept of frequency reuse in cellular communication. Apply it to show how it helps improve spectrum efficiency in a dense urban network. (5)
  - A)
  - B) Compare macro cells and micro cells by applying them to rural vs. city environments. (3)
  - C) Demonstrate how cell splitting improves capacity in a dense urban area. (2)

- 5) Illustrate the working principle of a pulse radar system using a block diagram, and analyze the role of the duplexer in signal transmission and reception. (5)
- A)
- B) Differentiate between the Plan Position Indicator (PPI) radar display and the Amplitude Scope (A-Scope) radar display. Explain how each format presents target range and direction information to the radar operator. (3)
- C) Classify radar beacons based on their function and demonstrate one real-life application. (2)

-----End-----

| Event Name               |                | Subject                                        |       |
|--------------------------|----------------|------------------------------------------------|-------|
| 058-END SEMSTER-ODD-2025 |                | ECE_4313:INTRODUCTION TO COMMUNICATION SYSTEMS |       |
| #                        | Course Outcome | Description                                    | Marks |
| 1                        | 1              | CO 1                                           | 10    |
| 2                        | 2              | CO 2                                           | 10    |
| 3                        | 3              | CO 3                                           | 10    |
| 4                        | 4              | CO 4                                           | 10    |
| 5                        | 5              | CO 5                                           | 10    |
| Total                    |                |                                                | 50    |

\* NA - CO Not Selected

| Event Name               |             | Subject                                        |           |
|--------------------------|-------------|------------------------------------------------|-----------|
| 058-END SEMSTER-ODD-2025 |             | ECE_4313:INTRODUCTION TO COMMUNICATION SYSTEMS |           |
| #                        | Level       | No of Questions                                | Marks (%) |
| 1                        | Remember    | 0                                              | 0(0%)     |
| 2                        | Understand  | 7                                              | 19(38%)   |
| 3                        | Application | 7                                              | 26(52%)   |
| 4                        | Analysis    | 1                                              | 5(10%)    |
| 5                        | Create      | 0                                              | 0(0%)     |
| 6                        | Evaluate    | 0                                              | 0(0%)     |
| Total                    |             | 15                                             | 50 (100%) |